higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MULTIMEDIA SERVICES
NQF LEVEL 4

20 NOVEMBER 2019

This marking guideline consists of 11 pages.

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SECTION A

QUESTION 1

1.1	1.1.1	D
	1.1.2	A
	1.1.3	C
	1.1.4	B
	1.1.5	B
	1.1.6	B
	1.1.7	C
	1.1.8	C
	1.1.9	D
	1.1.10	B
	1.1.11	C
	1.1.12	C
	1.1.13	C
	1.1.14	C
	1.1.15	A

(15 × 1) (15)

1.2.1	True
1.2.2	False
1.2.3	True
1.2.4	False
1.2.5	False
1.2.6	False
1.2.7	True
1.2.8	False
1.2.9	True
1.2.10	True

(10 × 1) (10)
[25]

TOTAL SECTION A: 25

SECTION B

QUESTION 2

2.1 2.1.1 Network errors can generally be categorised as problems with queuing, latency, and jitter. ✓ The data queue can affect network performance in several ways. It is often confused with connection speed, ✓ although pipe size doesn't affect the speed at which data travels between nodes.

2.1.2 Distance can have a dramatic impact on network performance, ✓ especially when the applications haven't been optimised. ✓ The maximum speed at which data can be forwarded is the speed of light, which is 186,000 miles per second or 186 miles per millisecond. This packet forwarding delay becomes significant when an enterprise is running an international client/server application because of the total distance the data need to travel.

2.1.3 Although network data transfer does not typically involve much direct memory usage, the amount of available memory can significantly affect the performance of other operating system components.✓ Inadequate system memory will cause the system to access the hard drive more often, leading to poor performance when reading or writing to disk.✓Some versions of Microsoft Windows have internal limits on how quickly✓ data can be read from or written to files based on the fixed size of the Windows 'paged pool' buffer.✓ (Any TWO)

2.1.4 Software firewalls and filters attempt to block✓or alter network traffic using the same operating system✓ and CPU resources as applications. ALL such software work by stopping each data packet, checking it against a set of rules, and then deciding what to do with that packet.✓ Depending on the amount of checking, this can greatly delay the packet and consume substantial CPU resources.✓ This can be problematic for moving data at high speeds when there are multiple layers, such as combining firewall, content filter, router, and network address translation software. To minimize problems, disable any components or rules that are not necessary and adjust others to allow MTP/IP traffic without filtering.✓ (Any TWO)

(4 × 2) (8)

2.2 DHCP is the abbreviation for 'dynamic host configuration protocol'. DHCP is a protocol which allows the server to allocate an IP Address to every client or computer on the network automatically.✓ This means that whenever a computer joins a DHCP server, it will automatically be given an IP address.✓ A DHCP server can provide a client with a lot more information depending on the requirement of the situation. Nowadays, most of the devices are configured in such a way that they automatically search for a DHCP server and ask it to allocate an IP address automatically.✓ The DHCP client requests an IP address from the DHCP server for a specific time, which is called a lease. That means that the client 'rents' an IP address from the DHCP server for a specific period✓. If the client wants to continue using the specific IP address, the client needs to re-assign the address by renewing the lease.✓ This will happen before expiration of the lease if the client is still on the network.✓ (Any 3 × 1) (3)

2.3

Desktop Operating Systems	Server Operating Systems
• running client applications faster,✓ like Office, Photoshop, Games✓ • easy access to web services,✓ like email, browsing, searching✓ • rich media services✓ • easy to use for non-expert users can work on a domain as a member (professional edition only - home edition cannot join a domain)	• unlimited user connections✓ • use of large amounts of memory ✓ • can act as web server, ✓ database server, e-mail server and other server-like roles✓ • optimised for network, instead of local application execution ✓ • extended management • extended fault tolerance to avoid downtimes • can hold a domain expensive

(Any 5 × 2) (10)

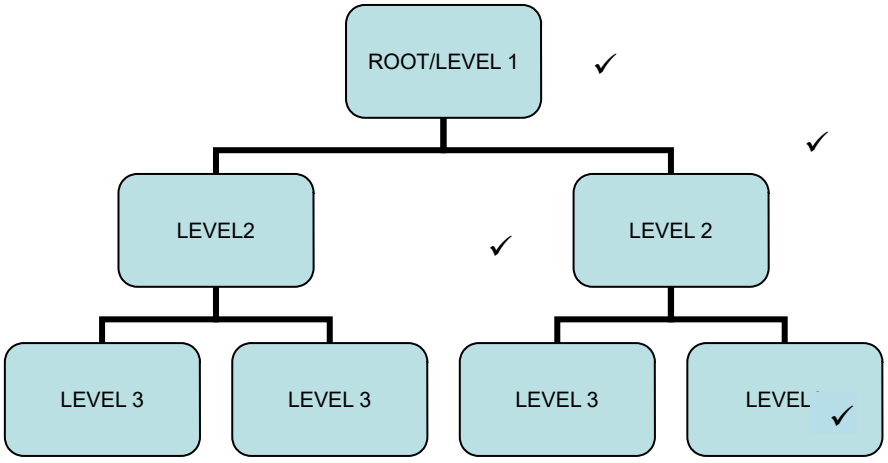
2.4
- Windows Server 2003✓
- Windows Server 2008✓
- Windows Server 2012✓
- Linux (excluding RHEL) ✓
- Windows Server 2000
- Red Hat Enterprise Linus (RHEL) (Any other applicable answers 4 × 1) (4)

[25]

QUESTION 3

3.1 Strong passwords cannot be hacked easily. Your server becomes fully secured.✓✓ The other benefits of strong passwords are that it can foil a brute force attack. (A brute force attack is a trial-and-error method used to obtain information such as a user password or personal identification number (PIN).✓✓ With strong passwords, you can save your server accounts from security hijacking.✓✓ Today, millions of employees work from home, attend virtual conferences and even complete college courses on websites protected by passwords. If it is strong, no unauthorized user can enter in it.✓✓ Passwords are never stored on the client computer. Empty passwords are not permitted by default (and they are strongly discouraged).✓✓
Simple to deploy, since the operating system provides the user accounts and password, almost no extra configuration is needed.✓✓ (Any 2 × 2) (4)

3.2



The hierarchical database structure looks like a pyramid. ✓ The record at the top is the highest level/ level1/root ✓ and the record at the bottom is the lowest level record✓. Each record can only be linked to one record above it, called the parent record, but can be linked to many records below it, called the child record. ✓
✓for at least 3 levels drawn
✓for each level
Accept any acceptable drawing depicting a hierarchical database structure (8)

3.3 3.3.1 SELECT * FROM Employee✓ WHERE✓ Age > 50 and City='Durban'✓ (3)

3.3.2 INSERT INTO Employee✓ (ID, First_Name, Last_Name, Address, City, Age)✓ VALUES✓ ('112', ' Bill', 'Baloyi', '20 Upper Rd', 'Brits', '48')✓ (4)

3.3.3 SELECT *✓ FROM Employee;✓
(Or any other applicable answer) (2)

3.3.4 DELETE FROM Employee✓ WHERE ID=102;✓
(Or any other applicable answer) (2)

3.3.5
- In this database only the ID column qualifies to be the primary key.✓
- A primary key is a special relational database table column (or combination of columns) designated to individually identify all table records.✓

A primary key's main features are:
- It should contain a unique value for each row of data.✓
- It cannot contain null values.✓
- A primary key is either an existing table column or a column that is specifically generated by the database according to a defined sequence.✓ ALL the other columns cannot uniquely identify the records of this table.✓ (Any 2 × 1) (2)

[25]

QUESTION 4

4.1 A traditional peer-to-peer network was a simple, inexpensive network✓ that connects less than 10 computers. ✓ Each computer called a peer, ✓ has equal responsibility and capacity. ✓ Ultimately the traditional peer to peer network was established to share resources and certain files✓ whereas internet Peer-to-peer or P2P is when the PC's are connected to the internet.✓ It allows users with the appropriate software and internet connection to copy files off of someone else's hard disk on the network. ✓ (7)

4.2 Web 2.0 is the name used to describe the second generation of the world wide web,✓ where it moved static HTML pages to a more interactive and dynamic web experience.✓ Web 2.0 is focused on the ability for people to work together and share information online via social media, blogging and Web-based communities.✓

Web 2.0 signalled a change in which the world wide web became an interactive experience between users and Web publishers,✓ rather than the one-way conversation that had previously existed.✓ It also represents a more accessible version of the Web, where new tools made it possible for nearly anyone to contribute, regardless of their technical knowledge.✓

(Any other applicable answer 4 × 1) (4)

4.3 function addTwoNumbers✓ (textBox1✓, textBox2✓)
{✓
var x✓ = document.getElementById(textBox1).value; ✓
var y✓ = document.getElementById(textBox2).value; ✓
var sum=0; ✓
sum✓=Number(x)+Number(y); ✓
alert("SUM is: " + sum); ✓
}✓
Any name can be used for the function
Any name can be used for the variables (12)

4.4 Internet freedom is a broad term covering several related concepts regarding use of the internet. These include, digital rights,✓ freedom of information,✓ right to internet access,✓ internet censorship✓ and net neutrality.✓ It is because of these that the internet has unleashed innovation, enabled growth, and inspired freedom✓ more rapidly and broadly than any other technological advance in human history. Its independence is its power✓. The internet offers a communications system uniquely free from government interference.✓ The internet is a crucial medium through which people can freely express their ideas✓ and share those ideas.✓ The internet has become an increasingly important tool through which democracy and human rights activists✓ mobilise and advocate for political, social, and economic reform.✓
(Any other applicable answer 2 × 1) (2)
[25]

QUESTION 5

5.1 5.1.1 By knowing the target audience your website will be catering to, the designer will try to reproduce the look and feel that associates with the above audience.✓ That is a worthy reason to get it right the very first time.✓
The very first step is to do some research.✓ If you want to create engaging content for your audience, you need to know what they need, and the only way to find out is by asking or reading about it.✓ (Any other applicable answers 2 × 1)

5.1.2 Based on the goals for your site✓ you can arrange the route that users should follow to move through the site. Strive to construct the site in such a way that the users can reach the places where they can accomplish a goal.✓
The web is primarily considered to be a means of communication, first to inform and second to promote.✓ Getting a message across is the most important thing it can do for a website.
(Any other applicable answers 2 × 1)

5.1.3 A sitemap is essentially an outline of the structure of the pages that will comprise the website.✓ Planning the sitemap prior to working on any website design has several benefits since you can build your design✓ around the most important pages, plan the website navigation more efficiently, and get an overall idea of the content that still needs to be written.✓ (Any other applicable answers 2 × 1)

5.1.4 It's essential that you know where your hard-earned money is going.✓ It can really help you to spend responsibly.✓ Don't allow someone else to make that decision for you. Grab the bull by the horns and steer it in the direction where you want your finances to go.✓ Budget forecasting can help you with that.✓
(Any other applicable answers 2 × 1)
(4 × 2) (8)

5.2 i is 20✓✓ (2)

5.3 5.3.1 NIDS is a system that attempts to detect hacking activities, denial of service attacks or port scans on a computer network✓ or a computer itself. The NIDS can monitor incoming, outgoing, and local traffic.✓
The NIDS monitors network traffic and helps to detect these malicious activities by identifying suspicious patterns in the incoming packets.✓ It actually resides on the Computer or the proxy for the LAN. ✓ (Any other applicable answers 3 × 1)

5.3.2 A host-based intrusion detection system (HIDS) is an intrusion detection system that is capable of monitoring✓ and analysing✓ the internals of a computing system as well as the network packets on its network interfaces,✓ similar to the way a network-based intrusion detection system (NIDS) operates.✓
It resides on the Computer or the proxy for the LAN.✓
(Any other applicable answers 3 × 1)

5.3.3 Stack-based IDS is latest technology, ✓ which works by integrating closely with the TCP/IP stack, ✓ allowing packets to be watched as they pass through the OSI layers.✓ Watching the packet in this way allows the IDS to pull the packet from the stack before the OS or application has a chance to process the packets. ✓
Usually placed in the router or switch for the network. ✓
(Any other applicable answers 3 × 1)
(3 × 3) (9)

5.4 5.4.1 Information leakage happens whenever a system that is designed to be closed to an observer reveals some information to unauthorised parties nonetheless.✓ Information leakage is an application weakness where an application reveals sensitive data,✓ such as technical details of the web application, environment, or user-specific data.✓ Sensitive data may be used by an attacker to abuse the target web application, its hosting network,✓ or its users.✓ (Any other applicable answers 2 × 1)

5.4.2 Man-in-the-middle attacks: Session between computers and remote web servers are given unique session IDs, which should stay private between the two parties;✓ however, an attacker can hijack the session by capturing the session ID and posing as the computer making a request, ✓ allowing them to log in as an unsuspecting user and gain access to unauthorized information on the web server.✓ Attackers can use a number of methods to steal the session ID, such as a cross-site scripting attack used to hijack session IDs.✓ An attacker can also opt to hijack the session to insert themselves between the requesting computer and the remote server, pretending to be the other party in the session.✓ This allows them to intercept information in both directions and is commonly called a man-in-the-middle attack.✓

(Any other applicable answers 2 × 1)

5.4.3 Attack by SQL code injection: SQL (pronounced 'sequel') stands for structured query language; it's a programming language used to communicate with databases.✓ Many of the servers that store critical data for websites and services use SQL to manage data in their databases.✓A SQL injection attack specifically targets this kind of server, using malicious code to get the server to reveal information it normally would not.✓ This is especially problematical if the server stores private customer information from the website, such as credit card numbers, usernames and passwords, or other personally identifiable information, which are tempting and worthwhile targets for attackers.✓ A SQL injection attack works by exploiting any one of the known SQL weaknesses that allow the SQL server to run malicious code.✓

(Any other applicable answers 2 × 1)

(3 × 2) (6)

[25]

QUESTION 6

6.1

```
<!DOCTYPE html>✓
<html>✓
  <head>
    <title>Contacts Form</title>✓
    <link rel='stylesheet' type='text/css' href='contact.css'/>✓
     </head>
  <body>
    <div id='header'><h1>Contact Us!</h1></div>✓
    <div id='wrapper'>
      <div id='inputs'>
        <form name='input' action='#' method='post'>✓
            <b> First name<b>: <input type='text' name='firstname' value='first
name'><br/>✓
              Last name: <input type='text' name='lastname' value='last
name'><br/>✓
                E-mail: <input type='email' name='email' value='e-mail
address'><br/><br/>✓
       </form>
      </div>
      <div>
       <form id='options'>✓
Message: <br/><textarea name='msg'></textarea><br/>✓
        <p>Sex:</p>✓
        <input type='radio' name='sex' value='male'>Male✓
        <input type='radio' name='sex' ✓value='female'>Female<br/><br/>✓
          <input type='checkbox' name='notify' value='updates' checked>I'd like
to receive notices when this website updates.<br/>✓
         <input type='checkbox' name='notify' value='services'>I'd like to receive
emails from your partners.<br/><br/> ✓
        <input type='submit' value='Submit'>✓
       </form>
      </div>
    </div>✓
  </body>
</html>✓
```

(Any 12 × 1)

(12)

6.2

```
#header {✓
  position: relative;✓
  top: -10px;
  background-color: #00A0B0;✓ OR yellow
  border-top-left-radius: 15px;
  border-top-right-radius: 15px;
  height: 40px; ✓
}
```

(Any 3 × 1) (3)

6.3 textarea {✓
vertical-align: top;✓
width: 300px;✓
color: red;✓ (Any 3 × 1) (3)

6.4
- In a separate file (external)
- At the top of a web page document (internal)
- Right next to the text it decorates (inline) (3)

6.5
- Google Chrome
- Mozilla Firefox
- Opera Web Browser
- Safari Web Browser
- Internet Explorer
- Slimjet Browser
- Maxthon
- SlimBrowser
- Netscape Browser
- UC Browser (Any other applicable answers 4 × 1) (4)

[25]

TOTAL SECTION B: 125
GRAND TOTAL: 150

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